

"All Roads Lead to ChatGPT": How Generative AI is Eroding Social Interactions and Student Learning Communities

Summary

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This paper investigates the social consequences of generative AI (genAI) adoption among undergraduate computing students, arguing that tools like ChatGPT are eroding peer interactions, mentorship networks, and learning communities in ways that have not been adequately examined. While prior research has documented genAI's cognitive and metacognitive effects—including benefits such as personalized explanations and debugging support, alongside risks like over-reliance and threats to academic integrity—the authors contend that the social dimensions of this technological shift remain largely unexplored. Their central thesis is that as students increasingly turn to genAI for help-seeking, the collaborative and communal structures essential to meaningful learning are breaking down.

The researchers conducted 17 semi-structured interviews with computing undergraduates (8 women, 9 men) across seven R1 universities in North America, employing reflexive thematic analysis. Participants varied in programming experience and genAI usage frequency, ranging from daily users who treat ChatGPT as their primary help resource to students who avoid it entirely. Several key findings emerged. First, 13 of 17 students described how peer help-seeking interactions are now mediated by genAI: when students ask classmates for assistance, they are frequently redirected to ChatGPT or given AI-generated outputs rather than authentic peer engagement. This phenomenon affected even non-users, who received AI-mediated help indirectly. Second, seven participants reported experiencing shame and stigma around genAI use, fearing judgment as "lazy" or "stupid," which created a paradoxical norm of private reliance and public hesitation. Third, students observed diminished diversity of perspectives and critical discourse, with one participant noting that AI provides only "Times New Roman" while peers offer distinctive "handwriting." Fourth, mentorship connections between senior and junior students were weakening, reducing access to the "hidden curriculum" of unwritten academic and professional norms. Fifth, while some students found genAI motivating for its efficiency, others reported demotivation and questioned the purpose of learning skills AI could perform. Finally, 11 of 17 participants noted increased feelings of isolation, with previously active online community spaces like class Discord servers falling silent as genAI supplanted peer question-and-answer exchanges.

The authors conclude that genAI may be inadvertently undermining the social fabric that supports student learning, belonging, and retention—outcomes with disproportionate consequences for underrepresented and first-generation students who benefit most from peer

support networks. They acknowledge limitations including the small, North America–focused sample and the anecdotal nature of the evidence, and they call for further empirical research. Their central recommendation is that educators must intentionally balance AI integration with strategies that foster and sustain peer interaction, mentorship, and community building, recognizing that the irreplaceable human dimensions of learning are at stake.